

5. Create visualizations that help evaluate strategy robustness:
 - Parameter heat maps showing performance across parameter combinations
 - Walk-forward performance charts comparing training vs. testing results
 - Drawdown profiles across different market regimes
 - Equity curves for each asset with trade markers
6. Produce a final report that includes:
 - Overall performance statistics across all walk-forward periods
 - Analysis of parameter stability and how it relates to performance
 - Comparison of strategy performance across different assets
 - Assessment of strategy robustness and potential for live trading

This project will give you hands-on experience building a professional-grade backtesting system that avoids common pitfalls like look-ahead bias and overfitting, while providing meaningful insights into strategy robustness.

Project 18: Multi-Method Optimization Comparison Framework

Objective: Create a system that implements and compares different optimization techniques to find robust strategy parameters while avoiding overfitting.